

Introduction to Research Methods & Experimental Design

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Welcome

Learning Outcomes

By the end of this lecture you will be able to:

- Explain what research methods are and why they matter
- Distinguish between everyday reasoning and scientific thinking
- Identify the key components of the research process
- Describe what you'll learn in this module

Who Are We?

Dr Gordon Wright

Module Coordinator

Research: Social cognition, deception, dark personalities and interpersonal perception

Office hours: Thursdays 12:00-13:00

Dr Bence Palfi

Lecturer in the department and our second Lab instructor

Research: AI, Human-Computer Interactions, Meta-Science

What Is This Module?

Research Methods & Experimental Design

How psychologists:

- Ask questions about the mind and behaviour
- Design studies to answer those questions
- Collect and analyse data
- Draw valid conclusions

Why Research Methods?

The Problem with Common Sense

- “Opposites attract”
- “Birds of a feather flock together”
- Both can’t be true... can they?

Common sense often gives contradictory answers. Science provides a systematic way to find out what’s actually true.

Everyday Thinking vs Scientific Thinking

Everyday Thinking

- Based on personal experience
- Influenced by biases
- Accepts anecdotes as evidence
- Confirmation bias
- “I just know”

Scientific Thinking

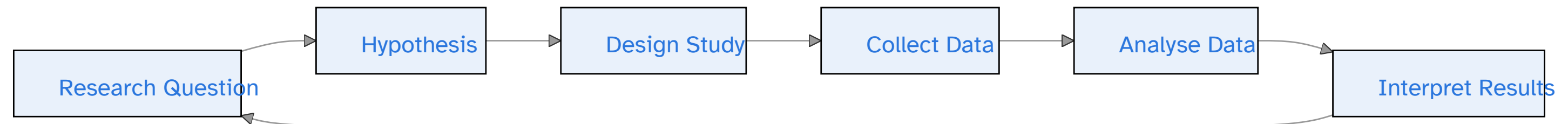
- Based on systematic observation
- Controls for biases
- Requires replicable evidence
- Seeks disconfirmation
- “Let’s test it”

Why Should You Care?

- Evaluate claims you encounter daily
- Understand how psychological knowledge is created
- Essential for your dissertation
- Transferable skills for any career
- Required for postgraduate study

The Research Process

From Question to Knowledge



Step 1: The Research Question

Where do research questions come from?

- Curiosity about behaviour
- Gaps in existing research
- Real-world problems
- Theory testing
- Unexpected observations

Example: Do people lie more when texting than face-to-face?

Step 2: The Hypothesis

A **hypothesis** is a specific, testable prediction about what you expect to find.

Good hypothesis:

“Participants will tell more lies per conversation when communicating via text message compared to face-to-face.”

Not a hypothesis:

“I wonder if texting affects lying.”

Step 3: Design the Study

Key decisions:

- Who will you study? (participants)
- What will you measure? (variables)
- How will you measure it? (methods)
- What comparisons will you make? (design)

We'll spend most of this module on this step.

Step 4: Collect Data

- Follow your plan precisely
- Treat participants ethically
- Record everything carefully
- Minimise errors and bias

Step 5: Analyse Data

Statistics are tools for finding patterns in data and determining whether those patterns are meaningful.

This is where **R** comes in — and what you'll learn in the labs.

Step 6: Interpret Results

- What do the findings mean?
- Do they support the hypothesis?
- What are the limitations?
- What questions remain?

Then the cycle continues...

What You'll Learn

Project Timeline

Week	Milestone
11	See how much fun collecting data is!
12-15	Develop research question and design
—	<i>Reading Week</i> — Work on Research Plan
16	Research Plan due ; Begin data collection
17-18	Data collection and initial analysis
19-20	Complete analysis; Develop write-up (term ends)
Spring Break	Individual report writing
Post-term	Research Report due Friday 1 May

Lectures vs Labs

Lectures (Wednesday 10-12 from week 12)

- Concepts and theory
- Research design principles
- How to think about data

Labs (Thursday 10-12)

- Working with real data
- Building practical skills
- Applying design principles
- Learning about statistical test with R

They work together — lecture concepts, lab application.

Why R? (nothing to fear!)

- Free and open source
- Industry standard in data science
- Powerful data analysis and visualisation
- Reproducible research
- Employable skill

No Experience Required

We assume you've never programmed before. We start from zero.

Assessment Overview

Find an [overview here](#)

Assessment	Weight	Deadline	Week
Research Plan	20%	Fri 27 Feb 2026, 12 noon	16
MCQ Exam	20%	Wed 11 Mar 2026, in lecture slot	18
EDS Assignment	20%	Fri 20 Mar 2026, 12 noon	20
Research Report	40%	Fri 1 May 2026, 12 noon	Post-term

Key Concepts

Variables

A **variable** is anything that can vary — any characteristic that can take different values across observations.

Examples:

- Age
- Reaction time
- Mood rating
- Experimental condition

Types of Variables

Categorical

- Distinct categories
- No inherent order (nominal)
- Or ordered (ordinal)

Examples: gender, condition, education level

Continuous

- Numerical values
- Can take any value in a range
- Interval or ratio scales

Examples: age, reaction time, score

Independent vs Dependent Variables

Independent Variable (IV): What the researcher manipulates or selects

Dependent Variable (DV): What the researcher measures as the outcome

Example: Testing whether caffeine affects reaction time

- IV: Caffeine (yes/no)
- DV: Reaction time (milliseconds)

Summary

Key Takeaways

1. Research methods provide systematic ways to answer questions about behaviour
2. Scientific thinking differs from everyday reasoning
3. The research process is cyclical: question → hypothesis → design → collect → analyse → interpret
4. Variables are the building blocks of research

Before Thursday's Lab

Preparation

1. Maybe read the intro [reading](#)
2. Bring curiosity!

No other preparation needed — we start from scratch.

Questions?

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See you in the lab on Thursday! Gordon & Bence